

## Features

- Low Gate Charge
- Low  $R_{DS(ON)}$ :
  - $24m\Omega$  @  $V_{GS} = 4.5V$
  - $28m\Omega$  @  $V_{GS} = 2.5V$
  - $34m\Omega$  @  $V_{GS} = 1.8V$
- Low Input/Output Leakage
- **ESD Protected up to 2kV HBM**
- **Lead Free By Design/RoHS Compliant (Note 3)**
- **Qualified to AEC-Q101 Standards for High Reliability**
- **"Green" Device (Note 4)**

## Mechanical Data

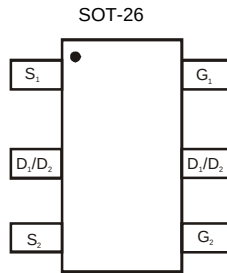
- Case: SOT-26
- Case Material - Molded Plastic, "Green" Molding Compound. UL Flammability Classification Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020D
- Terminals: Finish – Matte Tin annealed over Copper leadframe. Solderable per MIL-STD-202, Method 208
- Terminal Connections: See Diagram
- Marking Information: See Page 4
- Ordering Information: See Page 4
- Weight: 0.0008 grams (approximate)



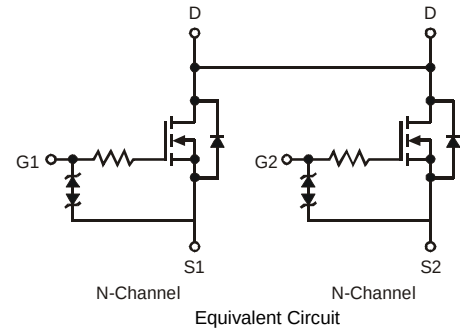
ESD PROTECTED TO 2kV



TOP VIEW



TOP VIEW  
Pin Configuration



N-Channel  
Equivalent Circuit

## Maximum Ratings @ $T_A = 25^\circ C$ unless otherwise specified

| Characteristic                    | Symbol   | Value    | Unit |
|-----------------------------------|----------|----------|------|
| Drain-Source Voltage              | $V_{DS}$ | 20       | V    |
| Gate-Source Voltage               | $V_{GS}$ | $\pm 12$ | V    |
| Drain Current (Note 1) Continuous | $I_D$    | 6.5      | A    |
|                                   |          | 5.2      | A    |
| Pulsed Drain Current (Note 2)     | $I_{DM}$ | 30       | A    |

## Thermal Characteristics @ $T_A = 25^\circ C$ unless otherwise specified

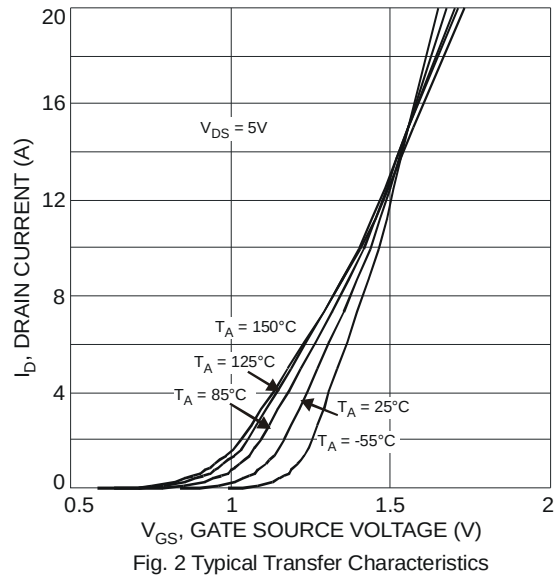
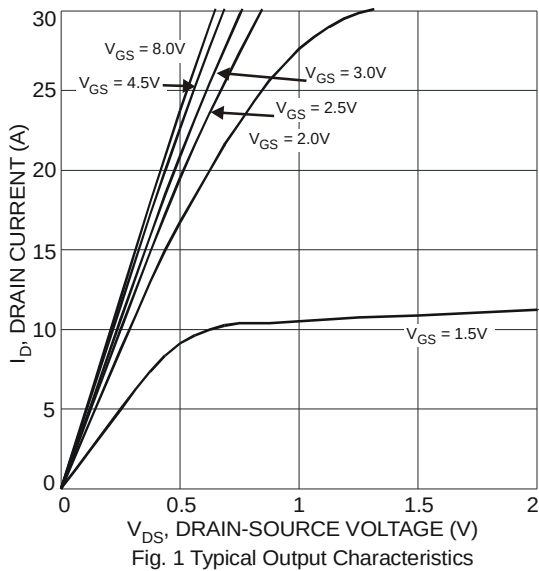
| Characteristic                                                | Symbol          | Value       | Unit         |
|---------------------------------------------------------------|-----------------|-------------|--------------|
| Total Power Dissipation (Note 1)                              | $P_D$           | 0.85        | W            |
| Thermal Resistance, Junction to Ambient (Note 1) $t \leq 10s$ | $R_{\theta JA}$ | 147         | $^\circ C/W$ |
| Operating and Storage Temperature Range                       | $T_J, T_{STG}$  | -55 to +150 | $^\circ C$   |

- Notes:
1. Device mounted on 1"x1", FR-4 PC board with 2 oz. Copper and test pulse width  $t \leq 10s$ .
  2. Repetitive Rating, pulse width limited by junction temperature.
  3. No purposefully added lead.
  4. Diodes Inc's "Green" policy can be found on our website at [http://www.diodes.com/products/lead\\_free/index.php](http://www.diodes.com/products/lead_free/index.php).

**Electrical Characteristics** @T<sub>A</sub> = 25°C unless otherwise specified

| Characteristic                             | Symbol               | Min | Typ | Max | Unit | Test Condition                                                                              |
|--------------------------------------------|----------------------|-----|-----|-----|------|---------------------------------------------------------------------------------------------|
| STATIC CHARACTERISTICS                     |                      |     |     |     |      |                                                                                             |
| Drain-Source Breakdown Voltage             | BV <sub>DSS</sub>    | 20  | —   | —   | V    | I <sub>D</sub> = 250μA, V <sub>GS</sub> = 0V                                                |
| Zero Gate Voltage Drain Current            | I <sub>DSS</sub>     | —   | —   | 1   | μA   | V <sub>DS</sub> = 20V, V <sub>GS</sub> = 0V                                                 |
| Gate-Body Leakage Current                  | I <sub>GSS</sub>     | —   | —   | ±10 | μA   | V <sub>DS</sub> = 0V, V <sub>GS</sub> = ±10V                                                |
| Gate-Source Breakdown Voltage              | BV <sub>SGS</sub>    | ±12 | -   | -   | V    | V <sub>DS</sub> = 0V, I <sub>G</sub> = ±250μA                                               |
| Gate Threshold Voltage                     | V <sub>GS(th)</sub>  | 0.5 | —   | 0.9 | V    | V <sub>DS</sub> = V <sub>GS</sub> , I <sub>D</sub> = 250μA                                  |
| Static Drain-Source On-Resistance (Note 5) | R <sub>DS (ON)</sub> | —   | 17  | 24  | mΩ   | V <sub>GS</sub> = 4.5V, I <sub>D</sub> = 6.5A                                               |
|                                            |                      |     | 20  | 28  |      | V <sub>GS</sub> = 2.5V, I <sub>D</sub> = 5.5A                                               |
|                                            |                      |     | 26  | 34  |      | V <sub>GS</sub> = 1.8V, I <sub>D</sub> = 3.5A                                               |
| Forward Transfer Admittance                | Y <sub>FS</sub>      | —   | 8   | —   | S    | V <sub>DS</sub> = 10V, I <sub>D</sub> = 5A                                                  |
| Diode Forward Voltage (Note 5)             | V <sub>SD</sub>      | —   | 0.7 | 1.0 | V    | I <sub>S</sub> = 2.25A, V <sub>GS</sub> = 0V                                                |
| DYNAMIC CHARACTERISTICS (Note 6)           |                      |     |     |     |      |                                                                                             |
| Input Capacitance                          | C <sub>iSS</sub>     | —   | 143 | —   | pF   | V <sub>DS</sub> = 10V, V <sub>GS</sub> = 0V<br>f = 1.0MHz                                   |
| Output Capacitance                         | C <sub>oSS</sub>     | —   | 74  | —   | pF   |                                                                                             |
| Reverse Transfer Capacitance               | C <sub>rSS</sub>     | —   | 29  | —   | pF   | V <sub>GS</sub> = 0V, V <sub>DS</sub> = 0V, f = 1MHz                                        |
| Gate Resisitance                           | R <sub>G</sub>       | —   | 202 | —   | Ω    |                                                                                             |
| SWITCHING CHARACTERISTICS                  |                      |     |     |     |      |                                                                                             |
| Total Gate Charge                          | Q <sub>g</sub>       | —   | 8.8 | —   | nC   | V <sub>GS</sub> = 4.5V, V <sub>DS</sub> = 10V, I <sub>D</sub> = 6.5A                        |
| Gate-Source Charge                         | Q <sub>gs</sub>      | —   | 1.4 | —   | nC   |                                                                                             |
| Gate-Drain Charge                          | Q <sub>gd</sub>      | —   | 3.0 | —   | nC   |                                                                                             |
| Turn-On Delay Time                         | t <sub>D(on)</sub>   | —   | 53  | —   | ns   | V <sub>DD</sub> = 10V, V <sub>GS</sub> = 4.5V,<br>R <sub>L</sub> = 10Ω, R <sub>G</sub> = 6Ω |
| Turn-On Rise Time                          | t <sub>r</sub>       | —   | 78  | —   | ns   |                                                                                             |
| Turn-Off Delay Time                        | t <sub>D(off)</sub>  | —   | 562 | —   | ns   |                                                                                             |
| Turn-Off Fall Time                         | t <sub>f</sub>       | —   | 234 | —   | ns   |                                                                                             |

Notes: 5. Test pulse width t = 300ms.  
6. Guaranteed by design. Not subject to production testing.



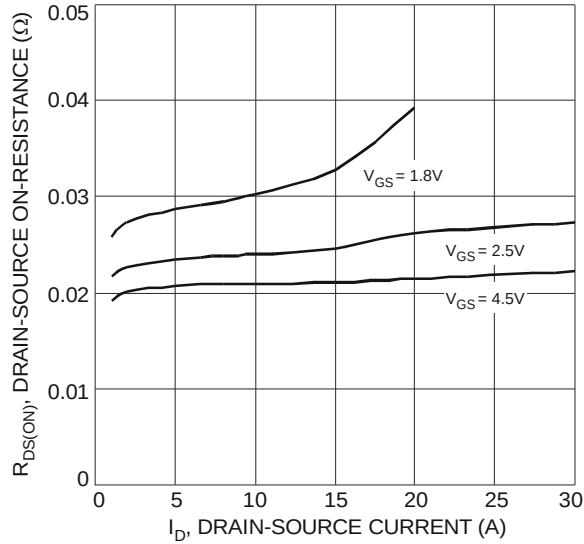


Fig. 3 Typical On-Resistance vs. Drain Current and Gate Voltage

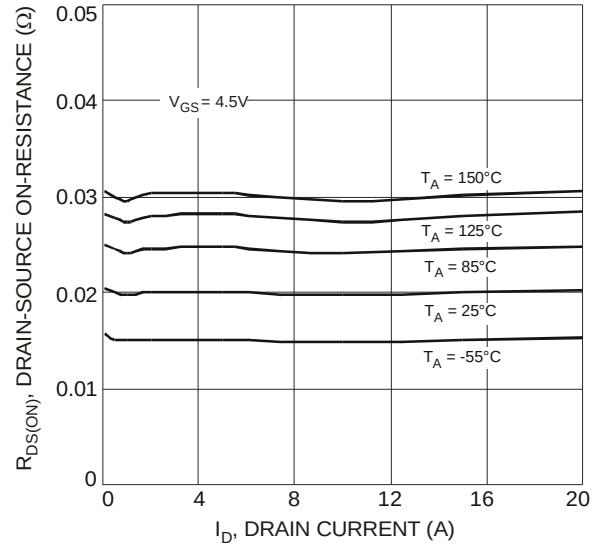


Fig. 4 Typical Drain-Source On-Resistance vs. Drain Current and Temperature

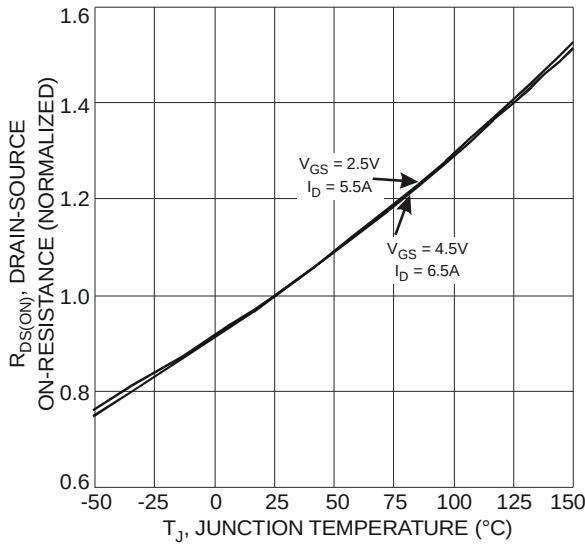


Fig. 5 On-Resistance Variation with Temperature

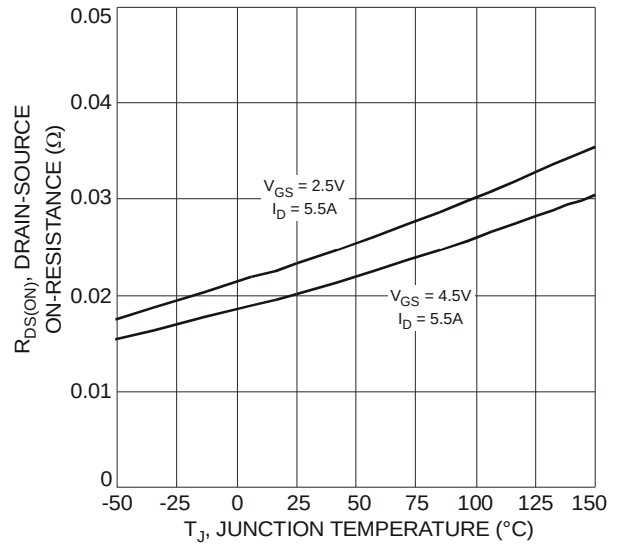


Fig. 6 On-Resistance Variation with Temperature

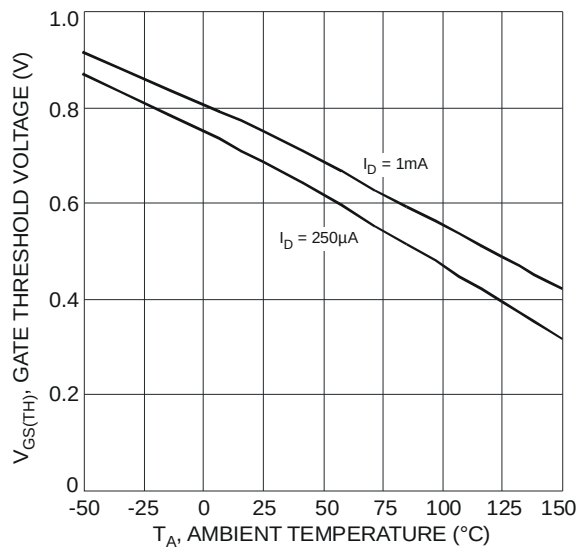


Fig. 7 Gate Threshold Variation vs. Ambient Temperature

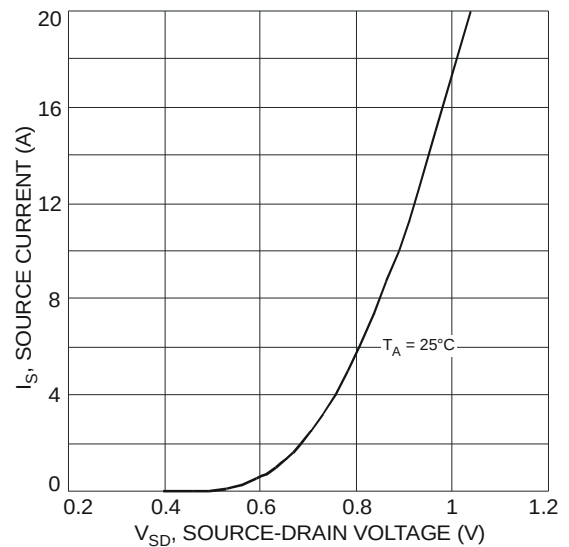
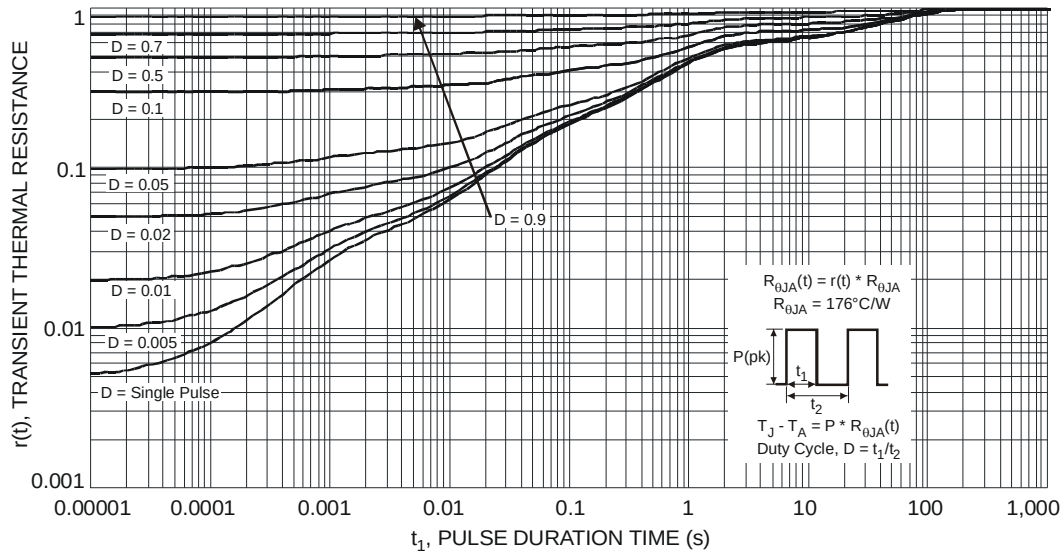
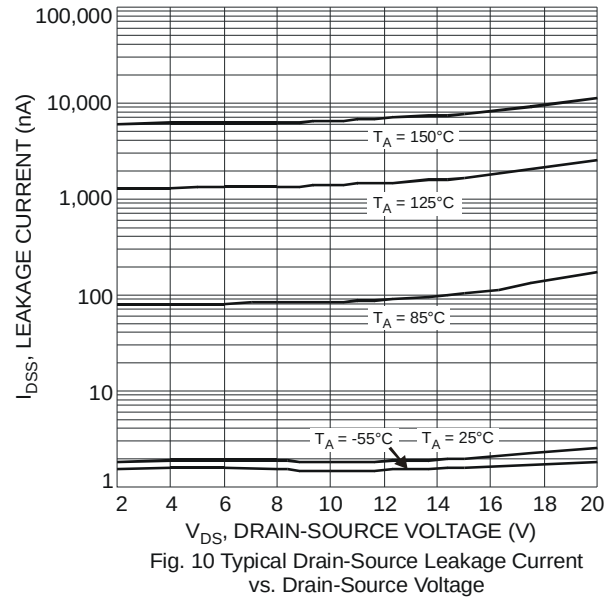
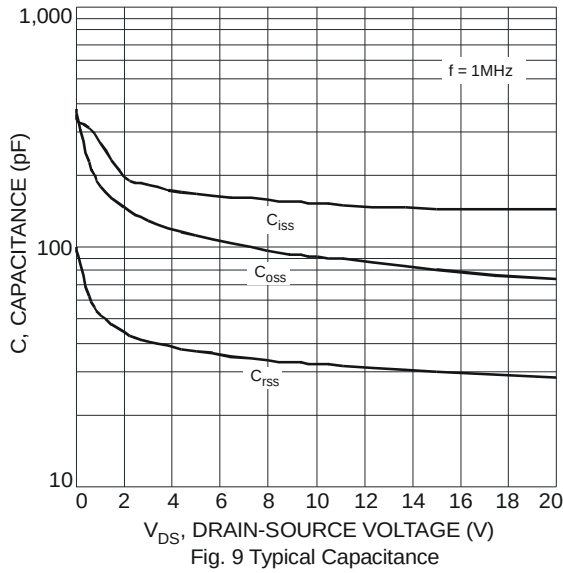


Fig. 8 Diode Forward Voltage vs. Current

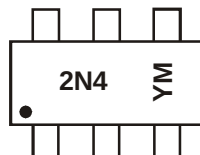


## Ordering Information (Note 7)

| Part Number  | Case   | Packaging        |
|--------------|--------|------------------|
| DMG6968UDM-7 | SOT-26 | 3000/Tape & Reel |

Notes: 7. For packaging details, go to our website at <http://www.diodes.com/datasheets/ap02007.pdf>.

## Marking Information



2N4 = Product Type Marking Code  
 YM = Date Code Marking  
 Y = Year (ex: W = 2009)  
 M = Month (ex: 9 = September)

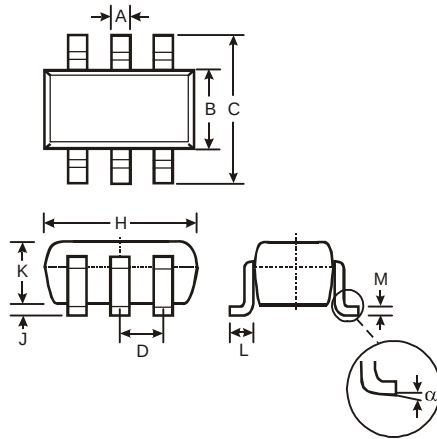
### Date Code Key

| Year | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
|------|------|------|------|------|------|------|------|------|
| Code | V    | W    | X    | Y    | Z    | A    | B    | C    |

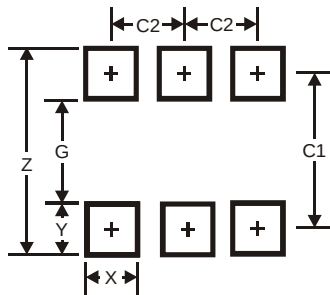
| Month | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Code  | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | O   | N   | D   |

# Package Outline Dimensions



| SOT-26               |       |      |      |
|----------------------|-------|------|------|
| Dim                  | Min   | Max  | Typ  |
| A                    | 0.35  | 0.50 | 0.38 |
| B                    | 1.50  | 1.70 | 1.60 |
| C                    | 2.70  | 3.00 | 2.80 |
| D                    | —     | —    | 0.95 |
| H                    | 2.90  | 3.10 | 3.00 |
| J                    | 0.013 | 0.10 | 0.05 |
| K                    | 1.00  | 1.30 | 1.10 |
| L                    | 0.35  | 0.55 | 0.40 |
| M                    | 0.10  | 0.20 | 0.15 |
| α                    | 0°    | 8°   | —    |
| All Dimensions in mm |       |      |      |

# Suggested Pad Layout



| Dimensions | Value (in mm) |
|------------|---------------|
| Z          | 3.20          |
| G          | 1.60          |
| X          | 0.55          |
| Y          | 0.80          |
| C1         | 2.40          |
| C2         | 0.95          |

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